

## RequestDispatcher Interface (Servlets)

### What is RequestDispatcher?

RequestDispatcher is an interface in the **Java Servlet API** used to **forward a request** from one resource to another or **include content** of another resource in the response.

Resources can be:

- Another servlet
- JSP page
- HTML file

### How to get RequestDispatcher

```
RequestDispatcher rd = request.getRequestDispatcher("resourceName");
```

### Main Methods of RequestDispatcher

#### 1. forward() Method

##### Definition

The forward() method **transfers the request** from one servlet to another resource.

##### Syntax

```
rd.forward(request, response);
```

##### Key Points

- Request is **forwarded internally** (client doesn't know)
- URL in browser **does not change**
- Only **one response** is sent
- After forwarding, **original servlet stops execution**

##### Example

```
RequestDispatcher rd = request.getRequestDispatcher("welcome");  
rd.forward(request, response);
```

👉 Control moves completely to "welcome" resource.

##### Use Case

- Login validation → forward to dashboard
- Form processing → forward to result page

## 2. include() Method

### Definition

The include() method **includes content** of another resource into the current response.

### Syntax

```
rd.include(request, response);
```

### Key Points

- Response from another resource is **added to current response**
- Original servlet **continues execution**
- URL **remains same**
- Multiple resources can contribute to response

### Example

```
PrintWriter out = response.getWriter();  
out.println("Header Content");
```

```
RequestDispatcher rd = request.getRequestDispatcher("footer");  
rd.include(request, response);
```

```
out.println("Main Content");
```

### Use Case

- Including header/footer
- Reusable components (like menus)

### Difference between forward() and include()

| Feature   | forward()             | include()                    |
|-----------|-----------------------|------------------------------|
| Control   | Transfers completely  | Returns back after execution |
| Response  | Replaced              | Appended                     |
| Execution | Stops current servlet | Continues                    |
| Use       | Navigation            | Content reuse                |
| Output    | One resource          | Multiple resources           |